



AI / MACHINE LEARNING

Interview questions you should know

LLM

50 questions

13

SECTIONS

50

QUESTIONS

Transformers · Attention · RAG · Fine-Tuning →



Akhila

AI & Tech



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SECTIONS 1 + 2

Q1-7

LLM Interview Questions

Questions 1–7 of 50



Section 1 - LLMs: Fundamentals & Comparison

1

What are Large Language Models (LLMs)?

2

How are LLMs different from traditional language models?

3

What are Foundation Models?

4

How is GPT-4 different from GPT-3 in capabilities and applications?

#

Section 2 - Tokenization & Embeddings

5

What is tokenization, and why is it important in LLMs?

6

What role do embeddings play in LLMs?

7

How do LLMs handle out-of-vocabulary (OOV) words?



Full answers & explanations

Check the link in the caption

Swipe →

Encoder-Decoder & Pretraining Objectives

Questions 8–12 of 50

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What are Sequence-to-Sequence (Seq2Seq) Models?

9

What is the difference between an encoder and a decoder?

10

How do autoregressive models differ from masked models?

11

What is masked language modeling, and how does it help pretraining?

12

What is Next Sentence Prediction (NSP)?



Full answers & explanations

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Transformer Architecture & Attention

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How does the Transformer architecture overcome challenges of traditional Seq2Seq models?

14

What are positional encodings in LLMs?

15

What is attention in Transformer models?

16

What is Multi-head attention?

17

How are attention scores computed in transformers?

18

What is the softmax function and why is it used in attention?

19

How is dot product used in self-attention?



Full answers & explanations

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Decoding & Sampling

Questions 20–23 of 50

- 20 What is beam search, and how does it differ from greedy decoding?
- 21 Explain the concept of temperature in LLM text generation.
- 22 What is the difference between top-k sampling and top-p sampling?
- 23 How does Adaptive Softmax speed up LLMs?



Full answers & explanations

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LLM Interview Questions

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Section 6 - Loss, Gradients & Backpropagation

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What is cross-entropy loss and why is it used in language models?

25

How are gradients computed with respect to embeddings?

26

What is the role of the Jacobian matrix in backpropagation?

27

What is the chain rule and how is it used in deep learning?

28

What is ReLU and why is it important?

29

What is the vanishing gradient problem, and how do Transformers solve it?

30

What are eigenvalues and eigenvectors (in dimensionality reduction)?

31

How is KL divergence used in evaluating LLMs?



Section 7 - Fine-Tuning & Parameter Efficiency

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What is LoRA and QLoRA?

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How can catastrophic forgetting be mitigated in LLMs?



Full answers & explanations

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Section 8 - Overfitting & Generative vs Discriminative

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What is overfitting, and how can it be prevented?

37

What are Generative and Discriminative models?

38

What is the difference between Discriminative AI and Generative AI?



Section 9 - Prompting

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How does prompt engineering influence the output of LLMs?

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What is Chain-of-Thought (CoT) prompting?



Full answers & explanations

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LLM Interview Questions

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Section 10 · RAG & External Knowledge

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What are the key steps in the Retrieval-Augmented Generation (RAG) pipeline?

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How does knowledge graph integration help LLMs?



Section 11 · Modern Architectures & Scaling

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How does Gemini's architecture improve training efficiency and stability compared to other multimodal LLMs?

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How does Mixture of Experts (MoE) improve scalability?



Section 12 · Learning Paradigms

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What is zero-shot learning in LLMs?

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What is few-shot learning in LLMs and why is it useful?



Full answers & explanations

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Deployment, Configuration & Challenges

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What is a context window?

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What is a hyperparameter?

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An LLM generates offensive or incorrect outputs. How do you handle it?

50

What are common challenges of using LLMs?



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